

## COVE ISLAND, ON, Canada

WMO: 714390

Lat: 45.327N

Lon: 81.735W

Elev: 181

StdP: 99.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.8      | -16.2 | -23.0                         | 0.5 | -18.1 | -20.5 | 0.6 | -15.2 | 16.4                  | -6.0 | 15.8 | -5.6 | 6.6                   | 320  | 0.658 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.8                    | 24.1            | 19.7 | 23.0 | 19.3 | 22.1 | 18.5 | 21.0                | 22.8 | 20.2 | 22.1 | 19.6 | 21.6 | 3.7                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.3                            | 15.3 | 22.3 | 19.5 | 14.6 | 21.7 | 18.7 | 13.8 | 21.0 | 61.8          | 22.9 | 59.1 | 22.2 | 56.6 | 21.6 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 14.9          | 13.2 | 11.7 | DB                         | -22.0 | 26.5               | 3.9 | 1.4                                                | -24.8 | 27.5       | -27.1 | 28.3       | -29.3 | 29.1       | -32.2 | 30.1 | (4) |
| (5)               |      |      | WB                         | -23.0 | 21.8               | 3.6 | 1.0                                                | -25.5 | 22.5       | -27.6 | 23.1       | -29.6 | 23.6       | -32.2 | 24.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.7         | -5.5  | -5.3  | -2.1  | 3.0   | 8.3   | 13.8  | 18.1  | 19.1  | 16.5  | 10.4  | 3.8   | -0.9  | (6)  |  |
|                                                                        | DBStd       | 9.51        | 5.67  | 5.38  | 4.51  | 3.31  | 3.17  | 3.32  | 2.54  | 2.27  | 3.36  | 3.77  | 4.08  | 4.76  | (7)  |  |
|                                                                        | HDD10.0     | 2147        | 480   | 429   | 376   | 212   | 74    | 5     | 0     | 0     | 1     | 43    | 189   | 338   | (8)  |  |
|                                                                        | HDD18.3     | 4356        | 738   | 663   | 634   | 461   | 312   | 141   | 36    | 18    | 73    | 247   | 437   | 597   | (9)  |  |
|                                                                        | CDD10.0     | 926         | 0     | 0     | 0     | 1     | 20    | 119   | 250   | 283   | 198   | 54    | 2     | 0     | (10) |  |
|                                                                        | CDD18.3     | 96          | 0     | 0     | 0     | 0     | 0     | 5     | 28    | 43    | 19    | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 73          | 0     | 0     | 0     | 0     | 1     | 10    | 24    | 32    | 7     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 3           | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.4         | 6.9   | 6.3   | 5.8   | 5.4   | 4.4   | 3.9   | 4.1   | 4.2   | 4.6   | 6.1   | 6.7   | 6.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 991         | 73    | 54    | 57    | 82    | 90    | 84    | 84    | 94    | 100   | 109   | 88    | 77    | (15) |  |
|                                                                        | PrecMax     | 1171        | 120   | 112   | 139   | 147   | 156   | 141   | 137   | 168   | 192   | 177   | 138   | 155   | (16) |  |
|                                                                        | PrecMin     | 811         | 38    | 20    | 8     | 31    | 33    | 29    | 38    | 47    | 38    | 48    | 41    | 33    | (17) |  |
|                                                                        | PrecStd     | 97          | 21    | 24    | 28    | 29    | 31    | 28    | 28    | 28    | 35    | 35    | 25    | 24    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 6.6   | 6.1   | 9.5   | 14.6  | 20.1  | 24.7  | 25.6  | 26.0  | 24.4  | 19.9  | 13.9  | 10.2  | (19) |  |
|                                                                        |             | MCWB        | 5.8   | 4.3   | 6.4   | 9.8   | 13.8  | 18.0  | 19.9  | 20.4  | 19.9  | 16.8  | 11.8  | 9.2   | (20) |  |
|                                                                        | 2%          | DB          | 4.3   | 3.9   | 6.8   | 11.3  | 16.8  | 22.1  | 23.8  | 24.1  | 22.8  | 17.9  | 12.0  | 7.7   | (21) |  |
|                                                                        |             | MCWB        | 3.3   | 2.1   | 4.8   | 8.0   | 12.3  | 16.9  | 19.3  | 20.5  | 19.3  | 15.4  | 10.3  | 6.3   | (22) |  |
|                                                                        | 5%          | DB          | 3.1   | 2.6   | 4.5   | 8.8   | 14.6  | 20.2  | 22.6  | 23.0  | 22.1  | 16.5  | 10.7  | 6.3   | (23) |  |
|                                                                        |             | MCWB        | 2.0   | 1.3   | 2.6   | 5.7   | 10.9  | 16.6  | 18.8  | 19.7  | 18.6  | 14.2  | 9.0   | 4.8   | (24) |  |
|                                                                        | 10%         | DB          | 1.9   | 1.5   | 2.9   | 6.9   | 12.8  | 18.7  | 21.6  | 22.2  | 21.6  | 15.2  | 9.4   | 5.1   | (25) |  |
|                                                                        |             | MCWB        | 1.0   | 0.4   | 1.3   | 3.9   | 9.5   | 15.5  | 18.4  | 18.7  | 18.4  | 12.7  | 7.8   | 3.6   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.0   | 5.5   | 7.3   | 11.8  | 16.0  | 19.7  | 21.7  | 22.2  | 21.4  | 17.5  | 12.1  | 9.5   | (27) |  |
|                                                                        |             | MCDB        | 6.5   | 5.9   | 8.9   | 13.2  | 18.4  | 22.8  | 23.8  | 24.2  | 23.0  | 19.0  | 13.3  | 10.2  | (28) |  |
|                                                                        | 2%          | WB          | 3.6   | 2.5   | 4.9   | 8.2   | 13.5  | 18.4  | 20.6  | 21.2  | 20.4  | 16.0  | 10.4  | 6.5   | (29) |  |
|                                                                        |             | MCDB        | 4.1   | 3.3   | 6.6   | 10.8  | 16.2  | 21.1  | 22.4  | 23.2  | 22.2  | 17.6  | 11.8  | 7.4   | (30) |  |
|                                                                        | 5%          | WB          | 2.2   | 1.4   | 2.9   | 5.9   | 11.4  | 17.2  | 19.7  | 20.5  | 19.6  | 14.5  | 9.2   | 5.0   | (31) |  |
|                                                                        |             | MCDB        | 2.9   | 2.4   | 4.1   | 8.3   | 13.8  | 19.5  | 21.7  | 22.4  | 21.6  | 16.1  | 10.5  | 5.9   | (32) |  |
|                                                                        | 10%         | WB          | 1.0   | 0.4   | 1.5   | 4.3   | 9.9   | 16.0  | 19.0  | 19.8  | 18.6  | 13.3  | 7.8   | 3.5   | (33) |  |
|                                                                        |             | MCDB        | 1.9   | 1.5   | 2.7   | 6.4   | 12.4  | 18.3  | 21.1  | 21.6  | 20.8  | 15.2  | 9.3   | 4.8   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.8   | 5.8   | 5.1   | 4.9   | 5.8   | 6.0   | 4.7   | 3.8   | 4.1   | 4.0   | 4.4   | 4.6   | (35) |  |
|                                                                        |             | MCDBR       | 5.4   | 5.0   | 6.2   | 8.5   | 9.3   | 8.2   | 6.1   | 5.3   | 4.5   | 5.4   | 5.4   | 4.7   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.5   | 4.9   | 4.6   | 5.6   | 6.1   | 4.7   | 3.5   | 3.3   | 3.3   | 4.3   | 5.3   | 4.9   | (37) |  |
|                                                                        |             | MCWBR       | 5.6   | 5.1   | 5.9   | 8.2   | 9.3   | 7.2   | 5.6   | 4.5   | 3.9   | 5.1   | 5.3   | 4.9   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.291 | 0.299 | 0.303 | 0.346 | 0.367 | 0.396 | 0.402 | 0.396 | 0.371 | 0.354 | 0.318 | 0.283 | (40) |  |
|                                                                        |             | taud        | 2.464 | 2.390 | 2.432 | 2.360 | 2.419 | 2.356 | 2.378 | 2.386 | 2.433 | 2.440 | 2.493 | 2.529 | (41) |  |
|                                                                        | Edn at Noon |             | 828   | 887   | 934   | 914   | 902   | 873   | 862   | 854   | 848   | 810   | 786   | 806   | (42) |  |
|                                                                        |             | Edh at Noon | 69    | 89    | 98    | 115   | 113   | 121   | 117   | 112   | 98    | 85    | 67    | 59    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.15  | 2.05  | 3.52  | 4.65  | 5.57  | 6.09  | 6.18  | 5.24  | 4.02  | 2.36  | 1.27  | 0.82  | (44) |  |
|                                                                        | RadStd      |             | 0.14  | 0.19  | 0.34  | 0.42  | 0.53  | 0.44  | 0.27  | 0.27  | 0.38  | 0.26  | 0.20  | 0.08  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (29 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page